

Worship Muguti
Cyber Operations

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1.0 Introduction

This report evaluates addresses operating system vulnerabilities, the spectrum of cyber operations, infrastructure management strategies, and the critical legal and ethical obligations the organisation must have.

2.0 Operating Systems, Network Attacks, and SOC Functionality

2.1 Operating System Vulnerabilities

Operating systems serve as the foundation of infrastructure. They have vulnerabilities based on their design. Vulnerabilities often come from code bugs, misconfigurations or architectural weaknesses that allow unauthorised privilege escalation.

When comparing Windows and Linux, the security features differs a lot. Windows is designed with a focus on usability and backward compatibility and relies heavily on a complex registry and a graphical user interface. This increases the attack surface and vulnerabilities often target the Registry or the Server Message Block protocol. Also Linux is modular and file system based. It is considered more secure due to its open source auditing and strict permission models which are root vs user. Linux vulnerabilities come from misconfigured permissions or outdated kernel modules. Security controls must be made to the OS and a one size fits all solution will fail if it does not address the specific architectural weaknesses of the OS in use. *Nivedhitha (2025)*

2.2 Types of Network Attacks

Network attacks are generally Denial of Service (DoS), Malware and Social Engineering.

- **Malware:** Malicious software encrypts data demanding payment for release eg Ransomware
- **Social Engineering:** Manipulating individuals into giving confidential information eg Phishing
- **DDoS:** Overwhelming a network with traffic to make it unusable.

Organisations must defend against these to ensure business safety and data integrity.

Fortinet 04/01/2026

2.3 SOC Functionality

A Security Operations Centre (SOC) functions as the central intelligence hub for an organisation's security.

A SOC uses Threat Intelligence Platforms to ingest data about external threats e.g new malware signatures and Intrusion Detection Systems to monitor internal network traffic for suspicious activities. However, a SOC cannot operate in isolation. It must be fully integrated into the organisation's broader security processes. *Kyndryl (02/01/2026)*

3.0 Cyber Operations: Offensive, Defensive & Intelligence-Based

3.1 Offensive Operations

Offensive cyber operations involve actively simulating attacks against an organisation's own systems. The goal is not to cause harm, but to identify exploitable vulnerabilities before malicious actors do. *UK defence (01/01/2026)*.

3.2 Defensive Operations

Defensive operations involve the protective measures and continuous monitoring required to secure an organisation. Organisations typically protect systems through a combination of:

- **Hardening:** Reducing the attack surface by disabling unnecessary services.
 - **Monitoring:** Using SIEM tools to log activity.
 - **Prevention:** Implementing firewalls and access controls to block unauthorised entry.
- UK defence (01/01/2026)*.

3.3 Intelligence Based Procedures

Cyber intelligence refers to the collection, analysis, and application of information regarding potential threats. It transforms raw data into actionable insight. Intelligence supports security decisions in two main ways which are proactive and reactive. *Cyber Threat intelligence (2025)*

4.0 Managing Information Infrastructure

4.1 System Management Areas

Managing information infrastructure requires an approach covering four concepts:

- **Deterrence:** Implementing controls that discourage attackers such as warning banners.
- **Detection:** The ability to identify a breach as it happens, utilising logs and monitoring tools.
- **Protection:** Mechanisms that stop the attack such as encryption, firewalls and antimalware software.
- **Reaction:** The processes activated after a breach is confirmed including containment and recovery strategies. *Atlassian 29/12/2025*

4.2 Key Security Components

Three components are vital for robust management:

- **Endpoint Protection:** Modern infrastructure is decentralised. Securing the endpoints is critical as they are often the entry point for attackers.
- **Network Segmentation:** By dividing the network into smaller subnetworks, organisations prevent an attacker from moving laterally across the entire system if one segment is breached.

- **Incident Response Planning:** A predefined plan ensures that when a breach occurs, the organisation acts calmly and effectively, minimising downtime and confusion. Faddom (2025)

4.3 Link to Operating System Vulnerabilities

Management strategies cannot be generic. They must consider the specific OS in use. For example, if InterGens utilises Windows, the management strategy must focus heavily on Active Directory security and rigorous "Patch Tuesday" updates. If using Linux, the strategy must prioritise the management of SSH keys and root privileges. The management framework must address the specific weaknesses inherent to the chosen operating system. Spektor, H. (2024)

5.0 Legal, Social, and Ethical Issues

5.1 Legal Considerations

Cybersecurity is governed by strict legal frameworks. Key laws include the **Data Protection Act 2018 / GDPR**. These laws control how personal data is stored and processed. Another law is the Computer Misuse Act 1990 which defines illegal access. InterGens PLC must adhere to these laws not only to avoid heavy fines but to operate lawfully within the UK market. lawsociety.org 2025

5.2 Social Considerations

Security incidents have a big social impact. A breach at InterGens could expose the personal data of employees or customers. This can lead to distress and identity theft. Beyond the immediate victims such incidents damage public trust. Companies must consider how they communicate during a crisis. Transparency is essential to maintaining the social contract between the business and the community. Chin, K. (2025)

5.3 Ethical Considerations

Organisations face ethical dilemmas in cybersecurity. For example, while monitoring employees is necessary for security, it raises privacy concerns. Organisations have a responsibility to handle data ethically. This can be done by collecting only what is needed and using it solely for security purposes, rather than for invasive surveillance. Chin, K. (2025)

5.4 SOC Responsibilities

The SOC is the guardian of the organisation's most sensitive logs and data. Consequently, SOC analysts are expected to maintain the highest ethical standards. They must ensure that the tools used for investigation are not abused to spy on colleagues or access data without authorisation. The SOC must operate within the legal framework, ensuring that evidence gathered during an investigation is admissible and that the privacy of innocent users is respected.

6.0 Conclusion

In conclusion, securing InterGens PLC requires a multi layered approach. It begins with understanding the specific vulnerabilities of the operating systems in use and deploying a SOC that integrates intelligence with active defense. Management must balance deterrence, protection, and reaction, while strictly adhering to legal and ethical standards to preserve public trust.

References

Nivedhitha (2025) Vulnerability Management: Process, Risk-Based Approach, and Best Practices.

Top 20 most common types of cyber attacks | Fortinet, 02/01/2026

Security Operations Center (SOC) | Kyndryl 01/01/2026

UK defence 04/01/2026

https://www.leidos.com/company/global/uk-europe/defence?gad_campaignid=21676115666&gad_source=1&gbraid=0AAAAA91Jod98e5pOkZ2DZcVtp5nsNWX3V

Cyber Threat intelligence (2025).
<https://www.nicybersecuritycentre.gov.uk/cyber-threat-intelligence>.

Atlassian 29/12/2026 What is IT infrastructure Management? | Atlassian.
<https://www.atlassian.com/itsm/it-operations/it-infrastructure-management#strategies-for-effective-it-infrastructure-management>.

Faddom (2024) Network Security Architecture: 8 key components and best practices.
<https://faddom.com/network-security-architecture-8-key-components-and-best-practices/>.

Spektor, H. (2024) Operating System Vulnerabilities: Understanding and Mitigating the risk.
<https://sternumiot.com/iot-blog/operating-system-vulnerabilities-understanding-and-mitigating-the-risk/>.

Cyber security guidance for solicitors 2025
<https://www.lawsociety.org.uk/topics/cybersecurity/cybersecurity-for-solicitors>.

Chin, K. (2025) Cybersecurity and social Responsibility: ethical considerations
<https://www.upguard.com/blog/cybersecurity-ethics>.